

this document studies asteroid impacts on earth by looking at each asteroid type and the effects it may cause, depending on its effect possible parameters

the parameters that are taken in consideration to create an impact (the inputs):

- _ Diameter
- _ Density (ρ) -> gives mass
- _ Velocity (v)
- _ Impact angle (θ)
- _ Impact location (land vs ocean)

generale workflow of the study

->first, the **impact energy(energie liberee)** :master variable that sets the scale for the initial explosion, crater size, ...(potential for all other effects) and the range of impact (e.g., Local, Continental, Global, Mass Extinction)

-> then the **diameter** : to understand the dimension of the object that transferring this energy so it consequences

->the study of the **strike angle** determine the efficiency and directionality of energy and material distribution

->at then end the **place where it hit**: effect on land / ocean

framework for asteroid impact, simulation and visualisation

1-energy(impact energy)

origines of impact energy

the effective impact energy — the one that combines all parameters (size, density, velocity, impact angle, and target) and directly tells us how much energy is delivered to the ground. That's the one used in impact cratering theory.

Impact Energy

The impact energy is the effective energy delivered to the surface by an asteroid, after accounting for impact angle and coupling efficiency.

$$E_{\text{imp}} = \frac{1}{2} \rho_{\text{imp}} \frac{4}{3} \pi \left(\frac{D}{2} \right)^3 v^2 \sin \theta \eta_{\text{coupling}}$$

Where:

- D = impactor diameter (m)
- ρ_{imp} = impactor density (kg/m³)
- v = velocity at impact (m/s)
- θ = impact angle from horizontal (rad or deg)
- η_{coupling} = coupling efficiency
 - Metallic (M-type): 0.8–1.0
 - Stony (S-type): 0.3–0.6
 - Carbonaceous (C-type): 0.1–0.3

This value, E_{imp} , represents the true energy budget that drives **all observable consequences** of an impact: crater excavation, blast overpressure, thermal radiation, ejecta, seismic waves, and (if ocean impact) tsunamis.

Ranging Distribution

whole spectrum: from asteroids that survive the atmosphere and just make local / regional impacts up to planet-scale collisions (the only ones that can truly re-shape Earth’s spin/orbit).

Impact Energy Ranges

Here are the main ranges of final impact energy E_{imp} and the corresponding scale of destruction:

Impact Energy Range	Level of Destruction
$10^{14} - 10^{16}$ J (≈ 0.02 – 2 Mt TNT)	Local: destruction limited to a city or metropolitan area
$10^{16} - 10^{18}$ J (≈ 2 – 200 Mt TNT)	Regional: devastation across a region or small country
$10^{18} - 10^{20}$ J (≈ 200 – 20,000 Mt TNT)	Continental: destruction across entire countries or multiple regions
$10^{20} - 10^{22}$ J (≈ 20,000 – 2,000,000 Mt TNT)	Global: worldwide climate disruption, possible mass extinction
$> 10^{22}$ J (> 2 million Mt TNT)	Planetary: Earth-wide catastrophe, potential tilt/orbit changes

Effect of Impact Angle on Impacts

The angle of impact (measured from the horizontal) affects both the **efficiency** of energy transfer into the ground and the **distribution** of destructive effects.

Although the impact energy formula already includes the angle through the $\sin\theta$ term (which reduces the total effective energy), the **pattern of damage** on the surface also changes significantly.

Steep Impact (60°–90°)

- **Energy coupling**: maximum; almost all of the asteroid's energy reaches the ground.
- **Crater**: circular, symmetric ejecta blanket.
- **Blast & thermal**: concentric and symmetric rings.
- **Interpretation**: most intense local destruction.

Moderate Impact (30°–60°)

- **Most common** impact angle range.
- **Energy coupling**: high, only slightly reduced compared to vertical.
- **Crater**: almost circular, small asymmetry in ejecta.
- **Blast & thermal**: nearly symmetric.
- **Interpretation**: usually taken as the **reference scenario** in studies.

Shallow Impact (15°–30°)

- **Energy coupling**: reduced; part of the energy dissipates in the atmosphere.
- **Crater**: smaller, often elongated; ejecta focused **downrange**.
- **Blast & thermal**: elongated footprint, with stronger effects **downrange**.
- **Extreme case**: if angle $<15^\circ$, the asteroid may **skip** the atmosphere instead of striking the ground.

Possible impact according to the ranges

Range:A - (City/Metropolitan Scale)

Scope: Minimal asteroid impacts. This is the smallest possible range of asteroids capable of reaching earth's surface without fully burning up in the atmosphere. Effects are limited to city scale, usually as an airburst shockwave or a very small crater, while smaller objects below this threshold disintegrate harmlessly before impact

1- Blast Overpressure (Peak ΔP at ground)

- **The Analogy:** The shockwave from a gigantic explosion. This is the force that physically knocks down buildings and trees.
- **Technical Specification:** This is the peak overpressure in kilopascals (kPa) or pounds per square inch (psi) at ground level.
 - **100 kPa (~14.5 psi): Severe Structural Damage.** Near-total building collapse. Radius: **0.5–1.5 km.**
 - **35 kPa (~5 psi): Heavy Damage.** Collapse of wood-frame buildings; serious injury threshold. Radius: **1–3 km.**
 - **10 kPa (~1.5 psi): Light Damage.** Widespread glass breakage. Radius: **5–15 km.**

2- Thermal Radiation (Integrated fluence at ground)

- **The Analogy:** The intense heat flash from a nuclear detonation. It instantly ignites materials and causes catastrophic burns.
- **Technical Specification:** This is the total radiant energy deposited per unit area in kilojoules per square meter (kJ/m²).
 - **100 kJ/m²: Robust Ignition.** Spontaneous ignition of wood, fabrics, and vegetation. Radius: **~1–2 km.**
 - **20 kJ/m²: 3rd-Degree Burns & Fires.** >99% probability of severe burns; ignition of thin fuels. Radius: **~1–3 km.**
 - **2-10 kJ/m²: Painful Burns & Spot Fires.** 2nd-degree burns; ignition of easily flammable materials. Radius: **~5–10 km.**

3- Cratering (if ground strike occurs)

- **The Analogy:** A large open-pit mine or a small reservoir suddenly appearing. This is the permanent scar of the impact.
- **Technical Specification:** The final crater dimensions and ejecta blanket distribution.

- **Final Diameter: 150–600 m** (upper outlier to ~800 m).
- **Depth: 10–70 m** (typically a 1:4 to 1:10 depth-to-diameter ratio).
- **Ejecta Thickness:** Decays rapidly from **tens of cm** at the rim to **mere mm** by **~2 km** distance.

4- Seismic Equivalent

- **The Analogy:** A noticeable, damaging earthquake centered directly on the impact point.
- **Technical Specification:** The energy is expressed as a moment magnitude (M_w) and the shaking is described on the Modified Mercalli Intensity (MMI) scale.
 - **Moment Magnitude (M_w): ~4.0–5.5** near source.
 - **Intensity (MMI): VI-VII** within **~5–15 km**. This represents moderate to strong shaking: felt by all, furniture moves, and light damage occurs.

5- Ocean Impact / Tsunami (if over water)

- **The Analogy:** A mega-tsunami generated by a sudden, massive displacement of water, like a submarine landslide.
- **Technical Specification:** Wave height is highly dependent on water depth and distance from the impact.
 - **Near-field Wave Height** (within 20–50 km): **~5–20 m**.
 - **Coastal Run-up:** Amplification upon hitting shorelines can easily exceed **>10 m**.
 - **Far-field Wave Height** (a few hundred km away): Typically decays to **≤0.5–1 m** due to geometric spreading and energy dissipation.

6- Timing of Effects

The sequence of events is critical for understanding the rapid onset of destruction. The initial thermal flash is instantaneous.

- **Shock Front Arrival:** The destructive blast wave travels at roughly the speed of sound (~340 m/s). For the radii in this range:
 - At the **glass breakage radius (5–15 km)**: Arrives **~15–45 seconds** after the flash.
 - At the **heavy damage radius (1–3 km)**: Arrives **~3–9 seconds** after the flash.
 - At the **severe damage radius (0.5–1.5 km)**: Arrives **~1–4 seconds** after the flash.
- **Ejecta Arrival:** The fallout of coarse debris follows ballistic trajectories.
 - Widespread coarse ejecta arrives within **~10 to 75 seconds** after the impact across the affected area.
- **Seismic Shaking:** Felt almost instantly (within **1-3 seconds**) across the entire zone due to the high speed of seismic waves.

Impact Angle Effects

impact consequences vary by angle while maintaining the same energy scale:

- **Steep Angles (60°-90°):** Maximum energy transfer creates symmetric circular damage patterns
- **Moderate Angles (30°-60°):** Balanced energy distribution with nearly circular effects
- **Shallow Angles (15°-30°):** Reduced ground energy creates elongated damage corridors

Examples:

- *Blast Zone:* Circular → Slightly oval → Elongated corridor
- *Crater Shape:* Perfect circle → Nearly circular → Oval elongated downrange
- *Damage Distribution:* Equal all around → Minimal directionality → Focused downrange

Range B - (Regional Scale)

Scope: Regional impacts. This energy range produces the same categories of effects as a smaller (city-scale) impact,(range A) but with dramatically increased intensity and geographic scale. The devastation can encompass an entire region or small country.

Blast Overpressure (*Scaled up from city-level*)

- **100 kPa (Total Destruction):** Radius expands to **2–5 km**. Reinforced concrete buildings collapse.
- **35 kPa (Heavy Damage):** Radius expands to **5–15 km**. Widespread structural failure and tree blowdown.
- **10 kPa (Glass Breakage):** Radius expands to **15–40 km**. Universal window breakage and light damage.

Thermal Radiation (*Scaled up from city-level*)

- **100 kJ/m² (Spontaneous Ignition):** Radius expands to **2–4 km**. Near-instant ignition of all combustibles.
- **20 kJ/m² (3rd-Degree Burns):** Radius expands to **10–20 km**. Causes severe burns and ignites wildfires.
- **2-10 kJ/m² (Painful Burns):** Radius expands to **20–50 km**. Causes secondary burns and spot fires over a vast area.

Crater Formation (*if ground strike; scaled up from city-level*)

- **Final Diameter: 1–5 km** (previously 150–600 m).
- **Depth: 100–500 m** (previously 10–70 m).
- **Ejecta Blanket:** Debris is several meters thick at the rim (previously tens of cm), burying the immediate landscape.

Seismic Effects (*Scaled up from city-level*)

- **Equivalent Magnitude: Mw ~6.0–7.0** (previously Mw ~4.0–5.5).
- **Shaking Intensity: MMI VII–VIII** (severe) within **~50–100 km** (previously within ~5–15 km). Capable of triggering regional landslides.

Ocean Impact / Tsunami (*if over water; scaled up from city-level*)

- **Near-field Wave Height: 50–100 m** (previously 5–20 m) within **50–100 km**.
- **Coastal Run-up: >50 m** possible (previously >10 m), causing catastrophic coastal flooding.
- **Far-field Waves: 5–10 m** (previously ≤0.5–1 m), representing a major trans-oceanic hazard.

Atmospheric / Environmental Effects *(New for this scale)*

- **Dust & Aerosols:** Significant material is injected into the stratosphere, a phenomenon not seen at lower energies.
- **Regional Climate Effects:** Can cause short-term regional cooling, darkness, and potential acid rain, extending the disaster beyond the immediate blast zone.

Timing of Effects *(Scaled up from city-level)*

- **Shock Front Arrival:** The destructive blast wave travels at roughly the speed of sound (~340 m/s). For the larger radii in this range:
 - At the **glass breakage radius (15–40 km)**: Arrives **~45–120 seconds** after the flash.
 - At the **heavy damage radius (5–15 km)**: Arrives **~15–45 seconds** after the flash.
 - At the **total destruction radius (2–5 km)**: Arrives **~6–15 seconds** after the flash.
- **Ejecta Arrival:** The fallout of coarse debris follows ballistic trajectories.
 - Widespread coarse ejecta arrives within **~1 to 4 minutes** after the impact across the affected region.
- **Seismic Shaking:** Felt almost instantly (within **1-10 seconds**) across the entire region due to the high speed of seismic waves.

Impact Angle Effects

impact consequences vary by angle while maintaining the same energy scale:

- **Steep Angles (60°-90°):** Maximum energy transfer creates symmetric circular regional devastation
- **Moderate Angles (30°-60°):** Balanced energy distribution with nearly circular regional effects
- **Shallow Angles (15°-30°):** Reduced ground energy creates elongated regional damage corridors

Range C (Continental Scale)

Scope: Continental impacts. This energy range represents a global catastrophe, with immediate effects devastating an area the size of a continent and secondary environmental effects threatening worldwide disruption. The energy release is equivalent to thousands to millions of the largest nuclear weapons.

1- Blast Overpressure

- **100 kPa (~14.5 psi):** radius **10–20 km** → complete destruction of all structures.
- **35 kPa (~5 psi):** radius **30–60 km** → major urban areas leveled.
- **10 kPa (~1.5 psi):** radius **100–200 km** → widespread glass break, structural damage.
- **Effect level:** An entire **nation-sized region** is within destructive blast zones.

2- Thermal Radiation

- **100 kJ/m²:** radius **10–15 km** → instant ignition of most fuels, unsurvivable burns.
- **20 kJ/m²:** radius **30–60 km** → widespread fires, 3rd-degree burns to exposed people.
- **2–10 kJ/m²:** radius **100–200 km** → ignition spots and mass casualties.
- **Effect level:** **Continental firestorms** possible; entire forests and cities set alight.

3- Crater Formation *(if ground strike)*

- **Final diameter:** **5–20 km**.
- **Depth:** **0.5–1.5 km**.
- **Ejecta blanket:**
 - Tens of m thick near rim.
 - 1–5 m thick within **10–20 km**.
 - **cm-scale** deposits out to **hundreds of km**.
- **Effect level:** Crater larger than major cities; ejecta fallout buries entire regions.

4- Seismic Effects

- Equivalent earthquake magnitude: **Mw ~7.0–8.0**.
- **Intensity (MMI):**
 - **VIII–IX** (severe to violent) within **100–300 km**.
 - Triggering of massive landslides and fault ruptures.
- **Effect level:** Comparable to the **largest historical earthquakes**, but focused at the impact site.

5- Ocean Impact / Tsunami (*if over water*)

- **Near-field waves:** **100–300 m** high within 100–200 km.
- **Coastal run-up:** >100 m possible on nearby coasts.
- **Far-field waves:** **10–30 m** across ocean basins; destructive trans-oceanic tsunami possible.
- **Effect level:** Multiple coastlines across oceans at risk.

6- Atmospheric / Environmental Effects

- **Dust and ejecta** injected into stratosphere (10^{13} – 10^{15} kg).
- **Sunlight reduction** for weeks–months at continental scale.
- **Regional cooling** and short-term climate disruption.
- **Ozone destruction** possible from NO_x chemistry in upper atmosphere.
- **Effect level:** Begins crossing from regional to **global environmental hazard**.

7- Timing of Effects (*Scaled up from regional-level*)

The sequence of events remains consistent but operates on dramatically extended timescales due to the continental-scale distances involved.

- **Shock Front Arrival:** The destructive blast wave travels at roughly the speed of sound (~340 m/s).
For the massive radii in this range:
 - At the **widespread damage radius (100–200 km)**: Arrives **~5–10 minutes** after the flash.
 - At the **urban destruction radius (30–60 km)**: Arrives **~1.5–3 minutes** after the flash.

- At the **total annihilation radius (10–20 km)**: Arrives **~30–60 seconds** after the flash.
- **Ejecta Arrival**: The fallout of coarse debris follows ballistic trajectories.
 - Widespread coarse ejecta arrives within **~5 to 15 minutes** after the impact across the continental-scale affected area.
- **Seismic Shaking**: Felt almost instantly (within **5-20 seconds**) across the entire continent due to the high speed of seismic waves.

8- Rotation / Orbit Effects

- **No measurable effect** on Earth's orbit or rotation.
- Energies still ~10 orders of magnitude below what is needed to perturb Earth's dynamics.

Impact Angle Effects

impact consequences vary by angle while maintaining continental-scale energy:

- **Steep Angles (60°-90°)**: Maximum crustal penetration creates deep circular craters with symmetric continental seismic waves
- **Moderate Angles (30°-60°)**: Balanced continental effects with standard crater formation and widespread damage
- **Shallow Angles (15°-30°)**: Elongated continental destruction corridors with one-sided ejecta distribution and directional tsunamis

Range D — (global scale)

Scope: Global catastrophe. Comparable to the Chicxulub impact (66 Myr ago) that caused mass extinction. Asteroid size typically 1–5 km (stony) at ~15–20 km/s.

1- Blast Overpressure

- **100 kPa (~14.5 psi):** radius **50–100 km** → complete destruction of all structures.
- **35 kPa (~5 psi):** radius **200–400 km** → heavy building and forest destruction.
- **10 kPa (~1.5 psi):** radius **500–1000 km** → widespread damage to infrastructure and vegetation.
- **Effect level:** Multiple countries simultaneously devastated.

2- Thermal Radiation

- **100 kJ/m²:** radius **30–50 km** → ignition of all combustibles, unsurvivable burns.
- **20 kJ/m²:** radius **200–400 km** → widespread fires, 3rd-degree burns.
- **2–10 kJ/m²:** radius **1000+ km** → spot ignition, severe burns.
- **Effect level:** Firestorms across continental scales. Secondary ignition from re-entering ejecta may trigger **global wildfires**.

3- Crater Formation

- **Final diameter:** **50–150 km** (Chicxulub: ~180 km).
- **Depth:** **3–5 km**.
- **Ejecta blanket:**
 - Hundreds of m thick near rim.
 - Tens of m thick at 50–100 km.
 - **cm–m deposits** spread globally.
- **Effect level:** Crater the size of small seas; ejecta veil covers the planet.

4- Seismic Effects

- Equivalent earthquake magnitude: **Mw ~9.0–10.0**.
- **Intensity (MMI):**
 - **IX–X** (violent to extreme) within **1000 km**.
 - Fault rupture, massive landslides globally possible.
- **Effect level:** Comparable to the **largest tectonic earthquakes ever recorded**, but triggered by impact.

5- Ocean Impact / Tsunami

- **Near-field waves:** **>500 m** high within 200–500 km.
- **Coastal run-up:** hundreds of m possible on adjacent coasts.
- **Far-field waves:** **10–100 m** trans-oceanic tsunamis; destruction across multiple ocean basins.
- **Effect level:** Entire ocean basins destabilized; global tsunami hazard.

6- Atmospheric & Global Environmental Effects

- **Dust & aerosols:** 10^{15} – 10^{17} kg injected into stratosphere.
- **Sunlight blocked:** 50–90% reduction for months to years.
- **Impact winter:** global surface cooling by **10–20°C** for 1–3 years.
- **Global fires:** ejecta re-entry ignites biomass worldwide → injection of soot/ CO_2 .
- **Acid rain:** vaporized rock + ocean salts produce nitric/sulfuric acids.
- **Effect level: Mass extinction driver:** collapse of food chains, long-term biosphere disruption.

7- Rotation / Orbit Effects

Rotation/Orbit quick rules (apply everywhere in Range D)

- **Orbit:** impact location **does not matter**. Earth ΔV is set by asteroid momentum direction; magnitude remains **negligible**.
- **Day length (ΔT):** scale your equatorial best-case by latitude: $\Delta T(\phi) \approx \Delta T_{\text{max}} \cdot \cos|\phi|$.
- **Tilt kick ($\Delta \epsilon$):** scale by $\sin|\phi|$ and by the component of momentum **perpendicular** to spin axis (max near poles, oblique approach).

Results for ΔT_{max} :

- **1 km:** ± 0.003 ms
- **3 km:** ± 0.080 ms
- **5 km:** ± 0.37 ms

Note: signs: **eastward (prograde)** → **shorter day**, **westward (retrograde)** → **longer day**. Near-vertical strikes give minimal ΔT regardless of latitude.

8- Timing

- **Shock front:**
 - Destruction radius (50–100 km): arrives 3–5 min
 - Heavy damage (200–400 km): 10–20 min
 - Glass breakage (500–1000 km): 25–50 min
- **Thermal radiation:** instant (at flash); global secondary heating from ejecta re-entry in hours.
- **Ejecta arrival:**
 - Local large fragments: 10–30 min
 - Global fallout: 0.5–3 h
- **Seismic shaking:** felt within seconds; Mw 9–10; global waves circle Earth in ~1 h.
- **Tsunami (ocean strike):**
 - Local megatsunami: minutes
 - Trans-oceanic waves: 6–12 h
- **Atmospheric effects:**
 - Sky darkening: hours
 - Sunlight reduction: days
 - Cooling (impact winter): weeks → years

Perfect! Here's **Range C** without the examples:

Impact Angle Effects

- **Steep Angles (60°-90°):** Maximum atmospheric coupling creates symmetric global climate disruption and even ejecta distribution
- **Moderate Angles (30°-60°):** Balanced global effects with standard atmospheric injection and widespread environmental collapse
- **Shallow Angles (15°-30°):** Hemispheric-biased climate effects with asymmetric ejecta distribution and directional atmospheric stripping

Impact Angle Effects on Range D (Global Scale)

- **Steep Angles (60°-90°):** Maximum atmospheric coupling creates symmetric global climate disruption and even ejecta distribution
- **Moderate Angles (30°-60°):** Balanced global effects with standard atmospheric injection and widespread environmental collapse

- **Shallow Angles (15°-30°):** Hemispheric-biased climate effects with asymmetric ejecta distribution and directional atmospheric stripping

Range E — Planetary Scale (>10²² J)

Scope: Impacts with energy exceeding 10²² Joules. This range encompasses a progression from the lower bound of **planetary-scale disruption** to the upper bound of **literal planet-altering events**. We subdivide this range to clarify the escalation of effects.

E1: Global Catastrophe / Near-Planet Killer (10²² – 10²³ J)

- **Impactor:** ~5 km to ~10 km diameter (stony) at typical velocities.
- **Concept:** This is the extreme upper end of a biospheric extinction event. It represents a more severe version of the Chicxulub impact, with darker, colder, and longer-lasting conditions that could push the biosphere to the brink of total collapse, with recovery taking tens of millions of years.

E2: Planet Killer (10²³ – 10²⁵ J)

- **Impactor:** ~10 km to ~50 km diameter.
- **Concept:** This is a true **planetary sterilization event**. Recovery of complex life is impossible. The surface environment is reset to a pre-biotic state for hundreds of millions of years. The energy is sufficient to cause global crustal melting and significant atmosphere loss.

E3: World-Former / Planetary Disruption (>10²⁵ J)

- **Impactor:** >50 km diameter, up to planetary embryos (Mars-sized, ~1000s of km).
- **Concept:** This is no longer an "impact" in the conventional sense; it's a **planetary accretion or disruption event**. It doesn't just kill the planet—it can remake it or destroy it. The Moon-forming impact (Theia) is the quintessential E3 event.

Table: Sub-Categories of Planetary-Scale Impacts (Range E)

Sub-Range	Energy (J)	Analog	Primary Effect	Planetary Mechanics?
E1	10 ²² – 10 ²³	Large Chicxulub+	Biosphere Annihilation	No measurable change

Sub-Range	Energy (J)	Analog	Primary Effect	Planetary Mechanics?
E2	10^{23} – 10^{25}	—	Planetary Sterilization	Yes. Changes to day, tilt.
E3	$>10^{25}$	Theia (Moon-Forming)	Planetary Remaking	Yes. Drastic alteration.

1- Blast Overpressure

Effect	E1: Global Catastrophe	E2: Planet Killer	E3: World-Former
100 kPa (Destruction)	Radius: 100–200 km	Radius: 500–1000 km	Hemispheric. The concept of a "blast radius" becomes meaningless as the entire hemisphere is scoured.
35 kPa (Heavy Damage)	Radius: 300–600 km	Radius: 1500–3000 km	Global. The shockwave circumnavigates the globe multiple times.
10 kPa (Glass Breakage)	Radius: 1000–2000 km	Near-global.	N/A. No structures remain.
Effect Level	Multiple continents devastated.	Destruction is truly global and simultaneous.	The solid earth itself is globally shocked and melted.

2- Thermal Radiation

Effect	E1: Global Catastrophe	E2: Planet Killer	E3: World-Former
100 kJ/m² (Ignition)	Radius: 50–100 km	Radius: 200–500 km	Global. The energy flux vaporizes the upper ocean layer and melts surface rock globally.

Effect	E1: Global Catastrophe	E2: Planet Killer	E3: World-Former
20 kJ/m ² (3rd-Degree Burns)	Radius: 200–500 km	Radius: 1000–2500 km	N/A.
2–10 kJ/m ² (Burns)	Global from re-entering ejecta.	Global.	N/A.
Effect Level	Guaranteed global firestorms.	Oceans boil locally; atmosphere heats significantly globally.	The impact flash itself provides enough energy to melt the crust. The atmosphere may be partially vaporized.

3- Crater Formation

Effect	E1: Global Catastrophe	E2: Planet Killer	E3: World-Former
Final Diameter	100–250 km	500–2000 km (Multi-ring basin)	>3000 km. No discrete crater. The event is accretional (adds mass) or disruptional (shatters the planet).
Depth	5–10 km	10–30 km (penetrating the crust)	The entire planet is reworked. A global magma ocean forms.
Ejecta	Global layer ~1–5 m thick.	Global layer 10–100 m thick. Secondary impacts create craters everywhere.	Mass exchange. Significant mass is ejected into space (to form a moon or rings) or is accreted from the impactor.
Effect Level	Crater is a dominant geological feature for eons.	The crater is a planetary-scale feature that remodels the crust.	The planet is destroyed or remade into a new planetary body.

4- Seismic Effects

Effect	E1: Global Catastrophe	E2: Planet Killer	E3: World-Former
Magnitude (Mw)	~10.0–10.5	~11.0–12.0	>12.0. The entire planet is liquefied. Seismic waves are irrelevant.
Shaking (MMI)	X (Extreme) across the entire planet.	>X. Shaking exceeds all scales. Crust is fractured globally.	N/A
Effect Level	The planet "rings like a bell" for months. Triggers global volcanism.	Global crustal fragmentation. Widespread mantle melting.	Whole-planet melting and differentiation.

5- Ocean Impact / Tsunami

Effect	E1: Global Catastrophe	E2: Planet Killer	E3: World-Former
Near-Field Waves	~1 km high.	Several km high.	Planetary-scale. Waves are limited only by the depth of the atmosphere.
Far-Field Waves	50–100 m trans-oceanic.	100–500 m trans-oceanic.	The ocean is globally vaporized and redistributed.
Effect Level	Every coastline on Earth is obliterated.	Entire ocean basins are emptied and refilled multiple times.	The existence of liquid water oceans is temporarily terminated.

6- Atmospheric & Environmental Effects

Effect	E1: Global Catastrophe	E2: Planet Killer	E3: World-Former
Dust/Aerosols	10^{16}–10^{17} kg injected. Sunlight reduced by >99% for 5-10 years .	10^{18}–10^{19} kg injected. Complete darkness for decades .	The primordial atmosphere is completely lost and must be outgassed again from the molten planet.
Impact Winter	Global cooling of 15–25°C .	Global cooling of >50°C . The surface freezes solid to great depths.	N/A
Other Effects	Complete ozone destruction. Acid rain for centuries.	Atmosphere is stripped away or converted to a toxic mixture of rock vapor.	Atmosphere Loss. The new atmosphere is formed from volcanic outgassing.
Effect Level	Total collapse of the photosynthetic base. Mass extinction of all large life.	Sterilization. All life, including extremophiles, is eradicated. The planet is dead.	Re-formation. The planet's environment is completely reset from scratch.

7- Rotation & Orbit Effects

Effect	E1: Global Catastrophe	E2: Planet Killer	E3: World-Former
Rotation (Day Length)	No measurable change. The effect ($\leq$ milliseconds) is lost in natural variability.	Measurable change. Day length alters by seconds to minutes .	Drastic change. Day length changed by hours . The Earth's spin can be dramatically slowed or reversed.
Obliquity (Tilt)	No change.	Change of 1-5 degrees. This would	Change of >10 degrees. This would completely

Effect	E1: Global Catastrophe	E2: Planet Killer	E3: World-Former
		significantly alter climate zones.	redefine Earth's seasons and climate.
Orbit	No change. Energy is ~10 orders of magnitude too small.	Minor detectable changes in eccentricity or inclination.	Major changes. The Earth's orbital path around the Sun is significantly modified.
Effect Level	The "planet" is fine; the "biosphere" is not.	The planet's fundamental mechanics are noticeably, permanently altered.	The event sets the new rotational and orbital parameters for the remade planet.

8- Timing of Effects

Effect Type	E1: Global Catastrophe	E2: Planet Killer	E3: World-Former
Blast Overpressure	• 100 kPa: 100-200 km radius • 35 kPa: 300-600 km radius • 10 kPa: 1000-2000 km radius	• 100 kPa: 500-1000 km radius • 35 kPa: 1500-3000 km radius • 10 kPa: Near-global	• Hemispheric scouring • Shockwave circumnavigates globe • No structures remain
Thermal Radiation	• Ignition: 50-100 km radius • Burns: 200-500 km radius • Global firestorms	• Ignition: 200-500 km radius • Burns: 1000-2500 km radius • Oceans boil locally	• Global rock vaporization • Ocean layer vaporized • Atmosphere heating
Crater Formation	• Diameter: 100-250 km • Depth: 5-10 km • Ejecta: 1-5 m global layer	• Diameter: 500-2000 km • Depth: 10-30 km • Ejecta: 10-100 m global layer	• No discrete crater • Global magma ocean • Planetary remaking

Effect Type	E1: Global Catastrophe	E2: Planet Killer	E3: World-Former
Seismic Effects	• Magnitude: Mw 10.0-10.5 • Intensity X globally • Planet "rings" for months	• Magnitude: Mw 11.0-12.0 • Crustal fragmentation • Mantle melting	• Magnitude: >Mw 12.0 • Whole-planet melting • Planetary differentiation
Ocean Impact / Tsunami	• Waves: ~1 km high • Trans-oceanic: 50-100 m • Coastlines obliterated	• Waves: Several km high • Trans-oceanic: 100-500 m • Ocean basins emptied	• Planetary-scale waves • Oceans vaporized • Water cycle terminated
Atmospheric & Environmental	• Dust: 10^{16}–10^{17} kg • Darkness: 5-10 years • Cooling: 15-25°C	• Dust: 10^{18}–10^{19} kg • Darkness: Decades • Cooling: >50°C	• Atmosphere lost • Re-outgassing required • Complete sterilization
Rotation & Orbit	• Day: No change • Tilt: No change • Orbit: No change	• Day: Seconds-minutes change • Tilt: 1-5° change • Orbit: Minor changes	• Day: Hours change • Tilt: >10° change • Orbit: Major modification

ENDING!